Electric shaRed mOBility hUbs Trial (ROBUST): Co-design Workshop 1, Letterkenny

James Reynolds and Brian McCann (ATU)

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Railway stations, and other transport nodes often provide for more than just the movement of people and goods - they might be places with other services and facilities, or provide a '3rd space' for community interaction. Along similar lines, recent research has emphasised the 'Place' function of road facilities as a vital dimension, alongside the more traditional focus in traffic engineering on movement and access to adjoining land uses.[1]

Shared electric mobility hubs are new approach that might support more sustainable travel choices, especially in contexts where providing high frequency transit services may be challenging such as in regional towns. This may help with transition pathways towards lower carbon-emitting transport patterns, without some of the challenges associated with other forms of shared and micro-mobility (e.g. charging fire risks, inappropriate parking[2]). Such hubs may allow households to shift towards more walking, cycling and transit use, including through facilitating occasional other trips that might otherwise be challenging (e.g. beyond walking/cycling distance, outside typical service times, carrying luggage), but this might only be realised if communities can find them, and are incentivised to use them, including as part of transitions away from petrol-powered private vehicles.

Four shared e-mobility hubs have been implemented in Letterkenny, Galway, Waterford and Dundrum, with including e-bikes, an e-cargo bike and shared electric cars as an inital offering, plus *power supply infrastructure* to support onsite recharging and *future uses*. These represent a considerable capital investment in mobility infrastructure and potential opportunity for further development. However, it is unclear how community use of these hubs might best be supported so as to improve transport sustainability[3].

The ROBUST research project seeks better understanding of the encouragement, educational and other 'non-(major)-infrastructure' interventions[4] that might support community use of hubs, 'expand the mobility offer', build partnerships with community and other stakeholders so as to maximise impacts of each hub's implementation.

To this end, ATU is hosting Co-design Workshops with key stakeholders. to gain bottom-up insights from practitioners, prior to further engagement with others (including the general public). This note

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[1] Alexa Delbosc et al., "American Complete Streets and Australian SmartRoads: What Can We Learn from Each Other?" *Transportation Research Record*, 2018.

Figure 1: E-bikes @ the Letterkenny hub

[2] Ze Zhang, Yuchen Guo, and Li Feng, "Externalities of Dockless Bicycle-Sharing Systems: Implications for Green Recovery of the Transportation Sector," *Economic Analysis and Policy* 76 (2022): 410–19; Beth Ferguson and Jordan Scott Blandino, "Shared Micromobility Vehicle Design and Safety," 2025.

[3] Perhaps such hubs can support existing local programs and activities (e.g. meals-on-wheels, sporting clubs, library or council staff and services)? Or they might allow for pop-ups or additional facilities (e.g. café, luggage storage, delivery services) that increase activity and therefore improve safety and security (i.e. through Jane Jacob's "eyes on the street").

[4] John Williams, Andy Clarke, and Kathleen McLaughlin, *Balancing Engineering, Education, Law Enforcement, and Encouragement* (Washington, DC, USA: National Bicycling; Walking Study, US Department of Transport, Federal Highway Administration, 1993).

provides details of the agenda[5].

[5] Objectives include: 1. Stakeholder engagement, and an opportunity for practitioners to try the hub out for themselves! 2. Increase coordination of efforts going forward; 3. Initial steps towards further co-design workshops, potentially involving community stakeholders, and increasing the mobility hub offer.

8:00am: meet at hub

The Letterkenny hub is at the Justice Walsh Road carpark. We will meet there, inspect the site and then use the hub's vehicles to reach ATU for morning coffee/tea.

Figure 2: Buildings and streetnetwork near Letterkenny hub, by walking distance (m)

10:00am: co-design workshop

The UK Design Council[6] Double Diamond design process suggests alternating converging and diverging steps of: Discovery; Definiton, Development and Delivery. The co-design workshop will spend roughly 30 minutes on each step, tackling issues such as: signage and linemarking; minor works and placemaking; community, business and council engagement and events; and others as discovered.

[6] "The Double Diamond," undated, `https://www.designcouncil.org.uk/our-resources/the-double-diamond/`.

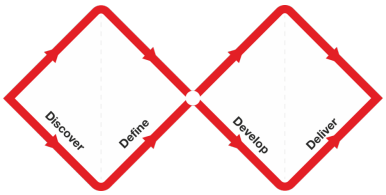

Figure 3: Double Diamond process

12 noon: return vehicles and close

. . . followed by lunch at McGettigan's Hotel

Details

- Facilitators: James Reynolds[7] and Brian McCann[8].
- Eight spaces available due to vehicle capacity limitations:
  1. Clodagh Campbell (Donegal Active Travel)
  2. Michael Canning (Donegal Active Travel);
  3. Lorraine Arbuckle (Climate Action Officer);
  4. Bryan Cannon (Donegal Council)
  5. Saqib Jamil (Donegal Council);
  6. Margaret Toner (Letterkenny Community Development Project);
  7. Mary Clyde (PPN Resource Worker);
  8. Martin McBride (North Donegal Community Network).
- Please bring a safety vest, sturdy shoes, a bicycle helmet and rain coat.

[7] `julianj.reynolds@atu.ie` or +353 87 348 3095. James is a post-doc researcher at ATU with more than 20 years of experience in transport engineering in Australia and Canada. He completed a PhD about legitimising transit prioritisation in 2020.

[8] `brian.mccann@atu.ie` Brian is a Lecturer in Civil Engineering at ATUD with extensive experience in research and teaching. He specialises in sustainable transport planning and policy, and transport infrastructure and regional connectivity.